

What SoCs comprise?

A system-on-chip consists of many sub-components such as one or more CPUs, graphics processing units (GPUs), digital, analogue, mixed-signal and radio frequency functions, random-access memory (RAM), read-only memory (ROM), Flash memory and EEPROM, oscillators and phase-locked loops (PLLs), real-time timers, analogue-to-digital converters (ADCs), digital-to-analogue converters (DACs), power management circuits and external interfaces such as USB, FireWire and Ethernet. All this is placed on a single substrate.

Simply put, the video encoding and decoding hardware powers the 'camcorder' functionality. The image processor ensures that photos are processed properly and saved quickly, and the audio processor frees the CPUs from having to work on audio signals thus allowing these to work on other tasks. Together, all these components and their associated drivers define the overall performance of a system.

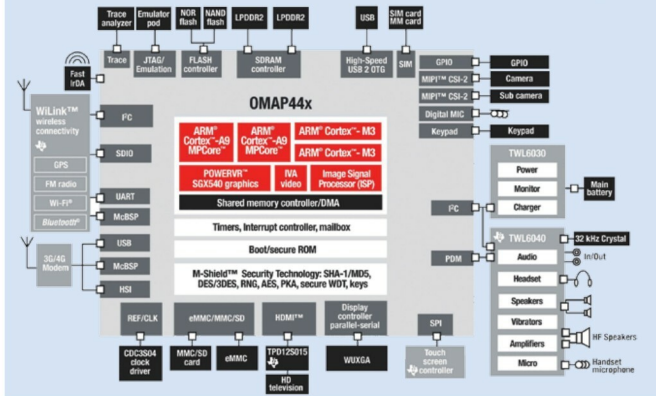


Fig. 1: Block diagram of an SoC (blocks highlighted in red are the critical blocks that define performance)

Fig. 2). It focuses on the CPU's processing power and does not utilise GPU's power for calculations. The test results are reported in numbers. The scores are susceptible to variation among devices with even the same SoCs.

BrowserMark appears to have a bug where one stage of their benchmark suite of tests could be intentionally skipped by using certain browsers. Apple A5 CPU benchmarks are hence higher than they should be.

GLBenchmark 2.1 Offscreen. This is an OpenGL ES benchmark with graphic scenes representing high-end gaming content (refer Fig. 3). The offscreen measurements are where all high-level scenes are rendered offscreen at 1280x720 pixel resolution. This method is claimed to provide apples-to-apples

performance comparison for all GPUs as it utilises the GPU's power as well for calculations.

Unlike the inflated BrowserMark score for the Apple A5, GLBenchmark score is accurate because the A5 does feature a high-end dual-core GPU.

Benchmark trends

If you look at the top three scoring SoCs (refer Fig. 2), you would find that clock speed does not define performance. Snapdragon MSM8260 despite claiming the highest clock speeds of 1.5 GHz still rates less than Apple A5 SoC running at 800 MHz (though there is a slight advantage due to the Safari-BrowserMark bug).

The reason why MSM8260 lost even with the spruced-up clock speed

is that this SoC is based on ARM's Cortex-A8 design. The other two SoCs that beat it are based on the Cortex-A9, which is known to perform 20 per cent better than A8. This result also shows why you should not compare performance of processors having different architecture based on clock speed alone. A better way to do so is to compare the performance per clock of the chips.

In the graphics benchmark (refer Fig. 3), Apple's A5 wins hands down with its PowerVR SGX543MP2 graphics solution. Obviously it's because this is a dual-core GPU. Moreover, SGX540 (belongs to Series 5) used in OMAP4460 was released around two years before the SGX543 (belongs to Series 5XT). As the chips used were